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# A palaeomagnetic and $^{40}Ar/^{39}Ar$ study of mafic dykes in southern Sweden: A new Early Neoproterozoic key-pole for the Baltic Shield and implications for Sveconorwegian and Grenville loops

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ABSTRACT

We present the results of palaeomagnetic and $^{40}Ar/^{39}Ar$ studies of the Proterozoic mafic dykes in the Norrköping and Falun areas of the southern Sweden. The primary remanence of two 939 ± 3 Ma dykes is supported by the rigorous baked contact test. The remanence direction of two other dykes, one of which was previously U–Pb dated at 946 ± 1 Ma is close to the reverse direction of 939 Ma dykes. Using these results together with previously published 935 ± 5 Ma palaeomagnetic data from the Göteborg-Slussen mafic dykes and some dykes from the Falun area we calculated the mean 946–935 Ma palaeopole for Baltica (0.9° S, 240.7° E, $A_{95}$ = 6.7), which can be qualified as the key pole. Using this pole together with other date we conclude that the Grenville and Sveconorwegian loops of Laurentian and Baltican Apparent Polar Wander Paths are temporary displaced by 100–150 m.y. We propose new palaeogeographic reconstructions of Baltica and Laurentia at ca. 940 Ma and ca. 850 Ma. We also present two new Mesoproterozoic non-key poles from 1410 Ma and 1595 Ma dykes.

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## 1. Introduction

Assembly of continents into a late Precambrian supercontinent, Rodinia, has been suggested in a number of studies (e.g. McMenamin and McMenamin, 1990; Dalziel, 1991, 1997; Hoffman, 1991). Models of Rodinia are mostly based on ca. 1350–1000 Ma orogenic belts, tectonic links between mobile belts across the cratonic margins, global analyses of the balance of Neoproterozoic and Early Palaeozoic continental margins and some palaeomagnetic data. Early Neoproterozoic tectonic reconstructions suggest a connection between Baltica and Laurentia (e.g. Hoffman, 1991; Dalziel, 1991, 1997; Weil et al., 1998; Dalziel et al., 2000; Pisarevsky et al., 2003; Li et al., 2008) in Rodinia. Since palaeomagnetism is the only tool for reconstructing the latitudinal position and orientation of cratons it plays an important role for these reconstructions. However, reliable palaeomagnetic data are sparse (e.g. Buchan et al., 2001) and more "key" palaeomagnetic poles, i.e. well dated poles that fulfill criteria for original magnetization, etc. (Van der Voo, 1990; Buchan et al., 2000) are needed for testing the palaeogeographic reconstructions. Due to the often high contents of magnetite, fast cooling and thereby small magnetic grain sizes and stable remanent magnetizations, mafic dykes are often excellent recorders of the Earth magnetic field. Palaeomagnetic data from mafic dykes forms a significant part in the palaeomagnetic databases (e.g. McElhinny and Lock, 1996; Pesonen et al., 2003; Pisarevsky, 2005).

In southern Sweden Early Neoproterozoic dyke swarms are widespread and have been the target for a large number of palaeomagnetic studies (e.g. Poorter, 1975; Patchett and Bylund, 1977; Bylund, 1985, 1992; Bylund and Elming, 1992; Bylund and Pisarevsky, 2002; Pisarevsky and Bylund, 1998, 2006). Most of these studies are from dolerite dykes in the Protogine Zone (PZ; Fig. 1) that defines a metamorphic boundary that separates the Southwest Scandinavian province in the west from the Transscandinavian Igneous Belt (TIB; Patchett et al., 1987) and the Southern Svecofennian subprovince (SSv) in the east (e.g. Gaál and Gorbatschev, 1987). The TIB and the Southwest Scandinavian province formed by crystal growth of the Baltic Shield with a westward younging during the TIB (ca. 1850–1670; Gorbatschev, 2004) and the Gothian period (1700–1550 Ma) (e.g. Åhäll and Larson, 2000; Andersen

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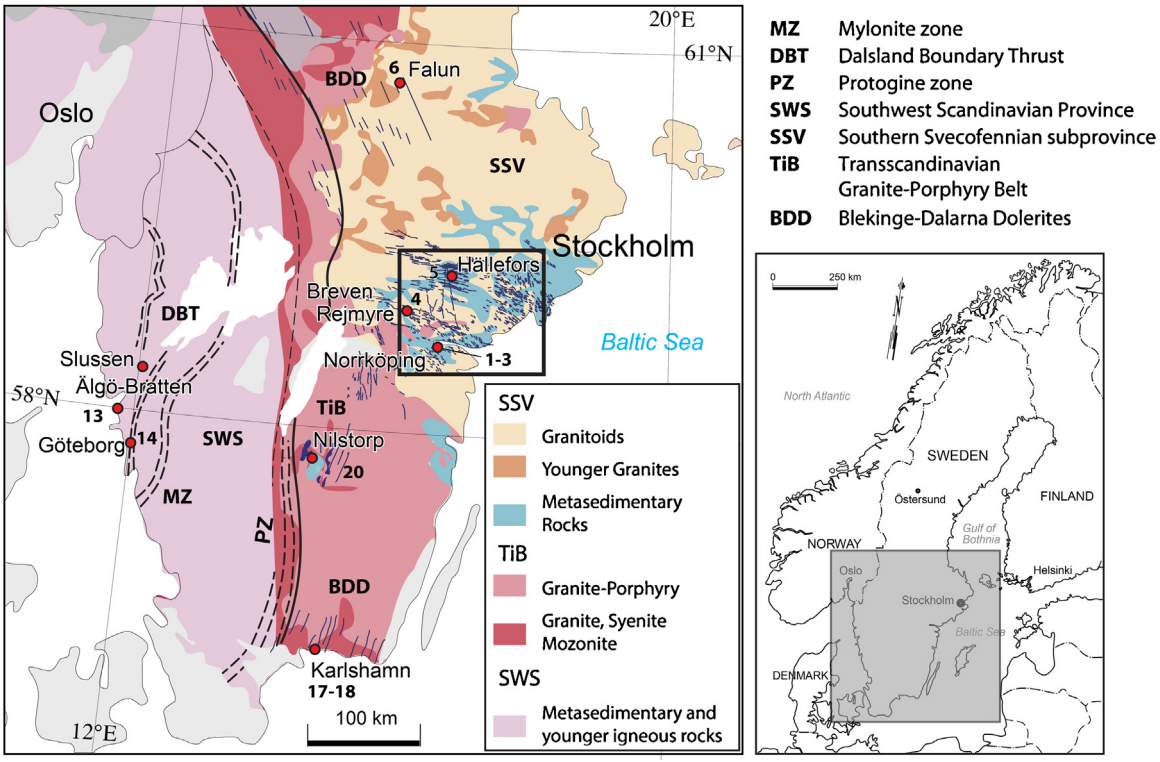



**Fig. 1.** The geology of the south-western part of the Fennoscandian Shield (a). The inlet in figure (a) marks the area where this study was concentrated on.

et al., 2004, Högdahl et al., 2004). From the late Palaeoproterozoic time the continental margin of a suggested continuous Laurentia-Baltica was the location of long lived subduction and terranes' accretion (e.g. Åhäll et al., 2000; Karlström et al., 2001; Andersen et al., 2004; Bingen et al., 2008; Pisarevsky and Bylund, 2010). The Sveconorwegian (in Laurentia Grenvillian; ca. 1150–900 Ma) continent-continent collision resulted in deformation and metamorphism of the Gothian orogen in Southwestern Sweden (e.g. Åhäll and Gower, 1997). The event resulted in a major deformation zone, the Mylonite Zone (Fig. 1; e.g. Gorbatschev, 1980), a zone that has been active during both Sveconorwegian thrusting and subsequent phases of exhumation (e.g. Stephens et al., 1996; Berglund, 1997). $^{40}Ar/^{39}Ar$ hornblende ages from the Mylonite Zone indicate deformations as late as ca. 915 Ma ago (Page et al., 1996), while ca. 40–70 km away from the Mylonite Zone two Sveconorwegian ages were identified, at 934–931 Ma and 1007–960 Ma, the younger ages suggested to be related to orogenic uplift. Deformation up to 40 km east of the Protogine Zone has been demonstrated from structural mapping (Wahlgren et al., 1994). Palaeomagnetic data older than ca. 900 Ma from southwestern Sweden should therefore be used with caution, since they may have been subjects to metamorphic overprints or tectonically disturbed and therefore not necessarily representing the Baltic Shield as a whole.

Palaeomagnetic results from 935 ± 3 Ma mafic dykes (Göteborg-Slussen dykes; Fig. 1) in the Southwestern Scandinavian province, west of the Mylonite Zone, were presented by Pisarevsky and Bylund (2006). The data are of good quality, however, no full field test demonstrating an original magnetization was presented. Here we will present a new ca. 935 Ma-highly reliable pole for the Baltic Shield. This is based on new palaeomagnetic data fulfilling field test of primary magnetization and $^{40}Ar/^{39}Ar$ age determinations from mafic dykes in the eastern part of the Southern Svecofennian subprovince. This pole is compared and combined with the palaeomagnetic data from the 935 Ma dykes in the Southwest Scandinavian province and 946 Ma dykes from Dalarna. The new combined palaeomagnetic pole forms an important input to the discussion of the shape and timing of the Sveconorwegian Apparent Polar Wander Path (APWP) loop (e.g. Patchett and Bylund, 1977; Elming et al., 1993), which is similar in shape to the Grenville Loop in the Laurentia APWP (e.g. Hyodo and Dunlop, 1993; Weil et al., 1998) and has been a basis for palaeogeographic reconstruction of Baltica vs. Laurentia at 935 Ma.

## 2. Geology, sampling and laboratory technique

Mafic dykes in the southeastern part of the Southern Svecofennian subprovince (Figs. 1 and 2) comprise several dyke swarms,

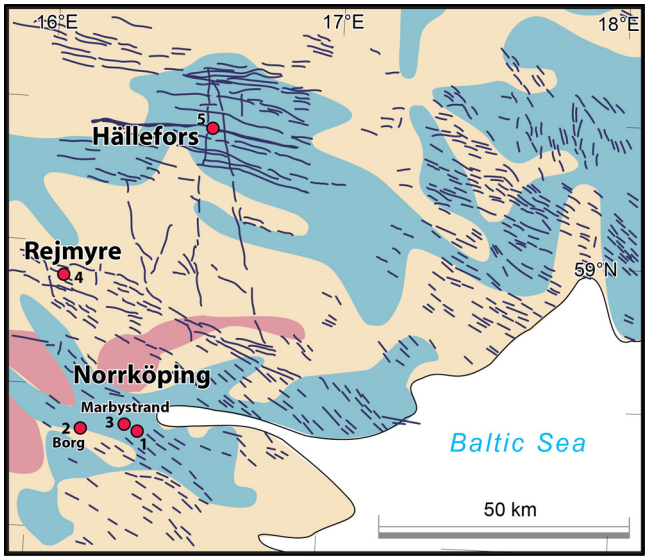



**Fig. 2.** Basic dykes in southern central Sweden (modified from Risku-Norja, 1992). Sampling localities for this study are marked by filled circles.

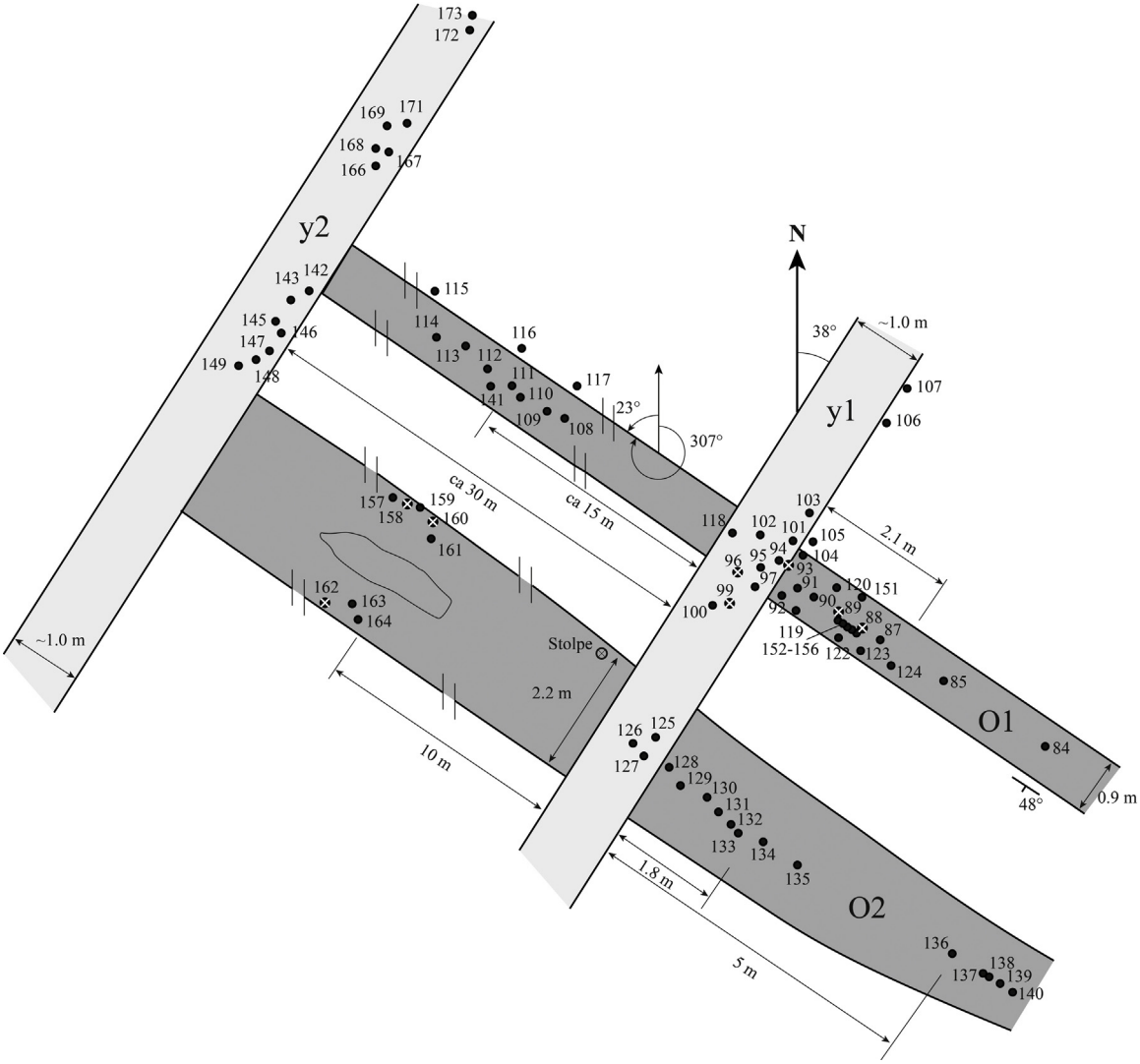



**Fig. 3.** The crossing dykes at site 1. The location of the collected samples for the palaeomagnetic analyses are marked by black dots and those also used for $^{40}Ar/^{39}Ar$ whole rock dating are marked by a white cross in the black dot. Close to the sample 88 in dyke O1 a number of samples (152–156) have been collected close to each other at increasing distance from the younger dyke. In dyke O2 a fragment of the host rock is included and marked by the thin black line.

predominantly trending in E–W directions, however, NW–SE, N–S and a few NE–SW trending dykes are also observed (Risku-Norja, 1992). The dykes intrude gneissic and granitic rocks that were formed during the Svecofennian orogeny at 1.9–1.8 Ga ago. In the gneisses there are often amphibolite layers that have been interpreted as intrusive sills from a late stage of the orogeny (Wikström, 1975). There are two major groups of 1.9–1.8 Ga intrusive rocks, a synorogenic suite of granitic to ultrabasic composition and a serorogenic suite with mainly migmatite gneisses. The rocks have been deformed and sometimes fractured.

The dykes are of tholeiitic to alkaline composition (Risku-Norja, 1992) and they were intruded after the cratonization following the Svecofennian orogeny, thus distinctly different from intraorogenic Svecofennian dykes also found in the region. Importantly the studied dykes are located relatively far from (>100 km), and to the east of the Protogine Zone and it is unlikely that they were influenced by late Sveconorwegian overprint. Trace elements data suggest that the dykes originate from a mantle source (Risku-Norja, 1992). E–W trending dykes, the Breven-Hällefors dykes (Bylund, 1985), have been dated at ca. 1595 ± 3 Ma (U–Pb, baddeleyite; Söderlund et al., 2005; Rejmyre, Fig. 2) and the N–S trending dykes probably belong to the subparallel Blekinge Dalarna Dolerites (BDD), which have been dated in the range 978–946 Ma (U–Pb baddeleyite; Söderlund et al., 2005). Apart from the Breven-Hällefors dykes no ages have been reported for dykes in the area of this study (Fig. 2).

In this study 97 samples were collected from 8 mafic dykes (Figs. 1 and 2; localities 1–5) in the southeastern part of the Southern Svecofennian subprovince and from one dyke (Falun, locality 6; Fig. 1) north of this area. In one locality (locality 1; Fig. 3) samples were collected from two NE (N38° E) trending dykes (Y1 and Y2) that cut two NW (N53° W) trending dykes (O1 and O2), which formed an excellent opportunity for baked contact tests. The younger Y1 and Y2 dykes are subvertical, while the O1 and O2 dip ca. 48° to the SW. Chilled margins are observed in both generations of dykes in the contact to the gneiss and in for the younger dykes also in contact to the older dyke. The samples were collected from various parts of the dykes and for the baked contact tests in the older dykes at increasing distances from the younger dykes.

Two other dykes, trending NW (localities 2 and 3, N50–60° W), were sampled to the west of locality 1, at a distance of 14 km and 2 km, respectively (Fig. 2). Both dykes are ca. 1 m wide and subvertical. The dykes at sites 4 (Rejmyre) and 5 (Hällefors) are E–W trending and belong to the Breven-Hällefors dyke system. The Rejmyre dyke is ca. 15 m wide and located ca. 40 km northwest of

**Table 1**
Summary of $^{39}Ar/^{40}Ar$ whole rock ages of the dykes at site 1.

| Sample | Unit | Run | Integrated age (Ma) | Plateau (P), Weighted average (W) in Ma | Fractions | $^{39}Ar$ release | Mean square weighted deviation |
|---|---|---|---|---|---|---|---|
| D96.99 | Y1 dyke | #1 | 959 ± 3 | 938 ± 4 (W) | 8 | 45 | 2.9 |
| D99.99 | Y1 dyke | #1 | 967 ± 3 | 941 ± 6 (W) | 10 | 46 | 4.4 |
| *Average*: | | | | 939 ± 3 | | | |
| D158.01 | O2 dyke | #1 | 1517 ± 4 | 1417 ± 9 (W) | 9 | 56 | 3.9 |
| D160.01 | O2 dyke | #1 | 1561 ± 5 | 1423 ± 6 (P) | 8 | 52 | 1.6 |
| D162.01 | O2 dyke | #1 | 1478 ± 5 | 1397 ± 6 (P) | 10 | 58 | 1.3 |
| *Average*: | | | | 1411 ± 9 | | | |
| D93.99 | Baked O1 dyke in contact with Y1 | #1 | 1300 ± 4 | 1172 ± 10 (W) | 4 | 34 | 3.8 |
| | | #2 | 1385 ± 4 | 1176 ± 14 (W) | 3 | 32 | 12.4 |
| *Average*: | | | | 1178 ± 8 | | | |
| D89.99 | O1 0.95 m from Y1 | #1 | 1354 ± 4 | 1385 ± 28 (W) | 7 | 66 | 38.0 |
| | | #2 | 1299 ± 4 | 1376 ± 13 (W) | 5 | 49 | 7.4 |
| *Average*: | | | | 1378 ± 12 | | | |
| D88.99 | O1 1.4 m from Y1 | #1 | 1355 ± 4 | 1428 ± 22 (W) | 4 | 49 | 26.1 |
| | | #2 | 1264 ± 4 | 1386 ± 15 (W) | 3 | 31 | 8.1 |
| *Average*: | | | | 1399 ± 20 | | | |

For samples from the baked contact and hybrid zone in the dykes O1 there are two runs to check for reproducibility. The weighted averages refer to the weighted average of plateau-like steps.

the locality 1. The rock at locality 5 forms a part the ca. 1 km wide Hällefors dolerite dyke and it is located ca. 60 km north–north–east of locality 1. Both dykes were sampled from their central parts. The last dyke included in this study (Falun, locality 6, Fig. 1) is situated north of the area described by Risku-Norja (1992). This dyke belongs to the 978–946 Ma old Blekinge Dalarna dolerite (BDD) dyke swarm (Solymon et al., 1992; Söderlund et al., 2005). In Dalarna different dykes in this swarm has previously been sampled for palaeomagnetic studies by Patchett and Bylund (1977), Bylund (1985) and Bylund and Elming (1992). Patchett and Bylund (1977) collected 10 samples at different locations along the same dyke as in locality 6 and here we collected 9 samples in a locality where that dyke has been dated at 945.7 ± 1.2 Ma (--Pb baddeleyite; Söderlund et al., 2005).

A portable rock coring drill was used for the sampling and the samples were oriented with sun and magnetic compasses.

The intensity and direction of remanent magnetization was measured with a SQUID (2G-DC) magnetometer and the specimens were AF-demagnetized in steps of 5 mT up to 30 mT and in 10 mT steps up to 120 mT. Thermal demagnetizations were performed using a Schonstedt (TDS1) oven at intervals of 50 °C up to 500 °C and 10–30 °C for higher temperatures. The remanence vectors were defined by least square fit of vectors in 3-D space (Torsvik, 1986; Torsvik et al., 1996) and by principle component analyses (Kirschvink, 1980). The magnetic susceptibility was measured on crushed specimens during heating and cooling using a κappabridge (KLY-3 and CS-3, AGICO).

Specimens from the palaeomagnetic samples were also submitted for $^{40}Ar/^{39}Ar$ whole rock dating at the Geochronology laboratory at the University of Alaska, Fairbanks. The cores were crushed, washed and sieved to obtain small (~0.5 mm) whole-rock chips for dating. The monitor mineral MMhb-1 (Samson and Alexander, 1987) with an age of 513.9 Ma (Lanphere and Dalrymple, 2000) was used to monitor neutron flux (and calculate the irradiation parameter, *J*). The samples and standards were wrapped in aluminum foil and loaded into aluminum cans of 2.5 cm diameter and 6 cm height. The samples were irradiated in position 5c of the uranium enriched research reactor of McMaster University in Hamilton, Ontario, Canada for 20 MW-h.

Upon their return from the reactor, the sample and monitors were loaded into 2 mm diameter holes in a copper tray that was then loaded in a ultra-high vacuum extraction line. The monitors were fused, and samples heated in a step-wise manner, using a 6-W argon-ion laser following the technique described in York et al. (1981), Layer et al. (1987) and Layer (2000). Argon purification was achieved using a liquid nitrogen cold trap and a SAES Zr–Al getter at 400 °C. The samples were analyzed in a VG-3600 mass spectrometer at the Geophysical Institute, University of Alaska Fairbanks. The argon isotopes measured were corrected for system blank and mass discrimination, as well as calcium, potassium and chlorine interference reactions following procedures outlined in McDougall and Harrison (1999). System blanks generally were $2 \times 10^{-16}$ mol $^{40}Ar$ and $2 \times 10^{-18}$ mol $^{36}Ar$, which are 10–50 times smaller than fraction volumes. Mass discrimination was monitored by running both calibrated air shots and a zero-age glass sample. These measurements were made on a weekly to monthly basis to check for changes in mass discrimination. Plateau ages (P in Table 1) were based on the criteria of three or more contiguous fractions (with a roughly constant Ca/K ratio) accounting for more than 50% gas release, having a Mean Square Weighted Deviation (MSWD) less than 2.5. For samples that show "plateau-like" segments, but not meet the criteria, a weighted average age of the plateau-like steps is calculated (W in Table 1). Some samples were run twice to check for reproducibility. A summary of all the $^{40}Ar/^{39}Ar$ results is given in Table 1, with all ages quoted to the ±1 sigma level and calculated using the constants of Steiger and Jaeger (1977). Integrated and plateau ages include error in the irradiation parameter (0.5%).

## 3. $^{40}Ar/^{39}Ar$ age determinations

Two samples from the "young" dyke Y1 (samples 96, 99; Fig. 3), three samples from the "old" dyke O2 (samples 158, 160, 162) and three samples from the contact (sample 93) and hybrid zone in dyke O2 (samples 88 and 89) were dated.

The age spectra from the samples collected in the dyke Y1 (96 and 99; Fig. 4) exhibit a slight amount of excess argon (anomalous old ages for the first steps) and do not meet the criteria for plateaus, but weighed mean ages for more than 45% of gas release are identical with an average age of 939 ± 3 Ma (Table 1). For sample 99, 7 fractions comprising 64% of gas release form a well-constrained isochron age of 941 ± 3 Ma (Fig. 4), confirming the weighted mean age 941 ± 6 Ma.

The samples from the O2 dyke, which were collected at ca. 15 m from the contact to younger dykes, show evidence of excess argon, especially in the high Ca/K steps. The average age from the three samples, two plateau ages and one weighted average age, is 1411 ± 9 Ma.

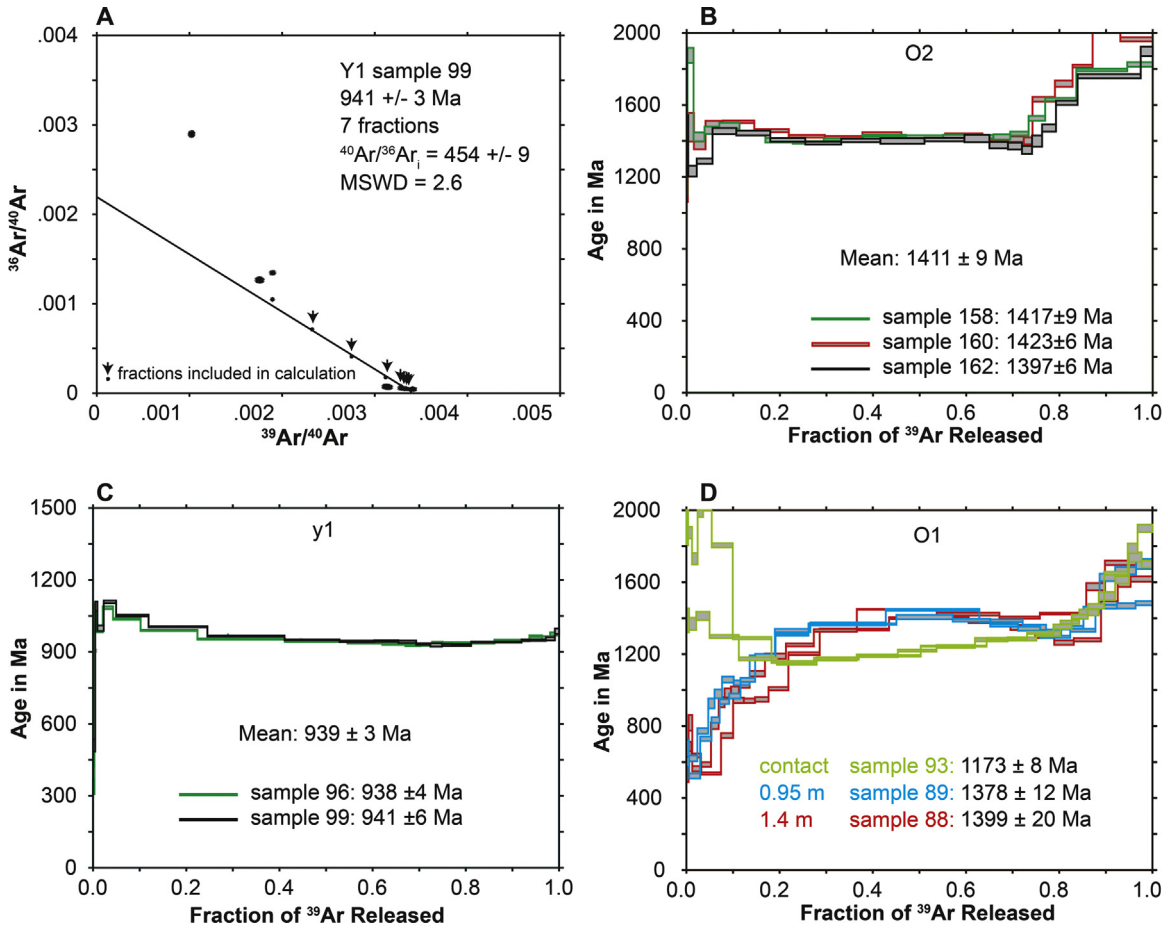



**Fig. 4.** $^{40}Ar/^{39}Ar$ whole rock age data (Table 1) from the dykes at locality 1 (for location of the samples see Fig. 3). The Y (O) denotes dated samples from the young (old) dykes. Some samples are run twice to check for reproducibility. For dyke Y1 an isochron age (A) was defined, with fraction included in the calculation shown by the small arrows. This age supports the weighted average age of the dyke (C). Ages from the dyke O1 are determined for samples ca. 15 m away from the young dykes (B) and for samples in the contact to the dyke Y1, and at 0.95 m and 1.4 m from the contact to Y1(D).

Sample 93 is collected from the baked contact in dyke O1 (Fig. 3) as part of a transect of samples collected at increasing distances from the contact to the young dyke Y1. Two runs of this sample give reproducible results with age spectra that are saddle-shaped, i.e. older at high and low $^{39}Ar$ release due to excess argon. The youngest weighted average age in each saddle is ca. 1173 Ma, which means the sample was not totally reset by the intrusion of the dyke Y1, but still retains some of its older argon. The spectra steps up to an age of ca. 1800 Ma, but a correct age for the dyke is difficult to estimate from these data. The samples collected at 0.95 m (89) and 1.4 m (88) from the contact were run twice and with reproducible results. The first part of the spectrum from the sample at 0.95 m is characterized by argon loss and very anomalous high Ca/K ratios, perhaps due to plagioclase microinclusions. The second part of the spectrum has low Ca/K ratios, reflecting the groundmass of the dyke, and an average age of 1378 ± 12 Ma. This age is about 30 Ma younger than the age of the O2 dyke as determined on samples outside the hybrid zone. That age difference could reflect either a different event or a small amount of argon loss due to the young dyke intrusion. The third part of the spectrum steps up to ages of ca. 1700 Ma and probably reflects excess argon. The spectrum from the sample 88 is more or less identical to that of 89 and the weighted average age from two runs is 1399 ± 20 Ma, i.e. similar to the age of the sample 89. The amount of excess argon in the high temperature release is much lower than that from the samples collected in the contact to the cutting dyke and at 0.95 m from the contact, suggesting a relationship between excess argon and distance from the dyke contact.

## 4. Palaeomagnetic results

### 4.1. Locality 1, crossing dykes

Samples were collected from the young dykes, the baked contacts and at increasing distance from the contact in the older dykes (O1 and O2). The characteristic remanent magnetization (ChRM) in the Y1 and Y2 dykes is isolated in a high coercivity (demagnetizing fields 25–120 mT) and medium to high unblocking temperature (400–580 °C; Fig. 5a) range and the mean directions are not significantly different (Y1: decl. = 311.2°, incl. = −43.3°, $\alpha_{95}$ = 5.7° and Y2: decl. = 313.3°, incl. = −40.7°, $\alpha_{95}$ = 7.4°; Table 2; Fig. 7a). The remanent magnetization in samples from the baked older dykes, within ca. 0.3 m from the contact, is isolated after treatment in temperatures between 500 and 580 °C (Fig. 5c). The mean direction (decl. = 311.5°, incl. = −45.4°; Table 2; Fig. 7a) is very well defined ($\alpha_{95}$ = 3.2°) and not significantly different from the ChRM of the cutting dykes Y1 and Y2. In the samples collected in the hybrid zone (see Elming et al., 2009), at distances of 0.8–1.7 m from the contact, two components of magnetization were isolated. The first one (in O1 and O2: decl. = 344.3°, incl. = −22.8°) is isolated in the Tub range 300–530 °C, while the second one (O1, decl. = 173.7°, incl. = 24.4°; O2, decl. = 179.9°, incl. = 10.8°) is defined in higher unblocking temperature (500–580 °C) ranges. Outside the hybrid zone, in intermediate to high unblocking temperatures (Tub ≥ 450 °C), remanences with shallow negative to positive inclinations are identified and also traced in very high coercivities (130–160 mT; Fig. 5b). The mean directions of

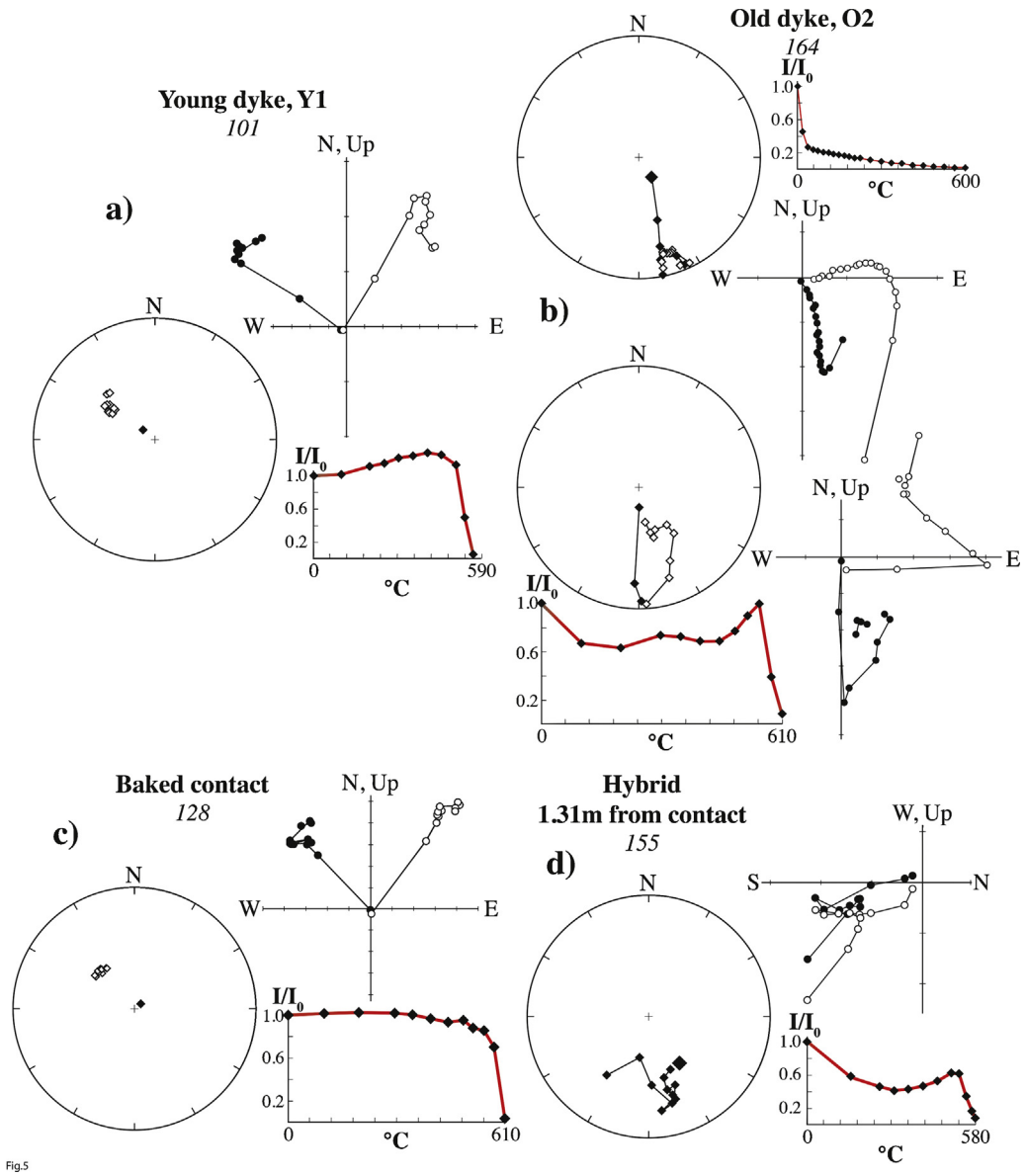



**Fig. 5.** Examples of remanence behavior during AF and thermal demagnetizations. In orthogonal plots, open (closed) symbols show magnetization vector endpoints in the vertical (horizontal) plane; curves show changes in intensity during demagnetization. Stereoplots show upwards (downwards) pointing palaeomagnetic directions with open (closed) symbols.

those remanences (O1: decl. = 168.5°, incl. = 16.9°; O2: decl = 169.9°, incl = 23.0°, Table 2) are not significantly different from the high Tub and coercivity components inside the hybrid zone. The high Tub-directions from the hybrid zone (H) and high coercivity and intermediate to high Tub-direction outside the hybrid zone (U) are combined for mean calculations (O1a (H + U); decl. = 172.6°, incl. = 22.9°; and O2b (H + U); decl. = 172.8°, incl. = 19.7°; Table 2; Fig. 7b). This exemplifies the "classical" positive contact test (Everitt and Clegg, 1962), i.e. distinct stable remanence directions in the cutting dyke, and two remanence components in the hybrid zone, one of which with a lower Tub similar to the ChRM of the cutting dyke and another with a higher Tub similar to the host' ChRM.

Outside the hybrid zone in one of the old dykes (O2) a low Tub and intermediate to high coercivity component is isolated with a direction (O2; decl. = 153.8°, incl. = −29.5°; Fig. 5b; Table 2) not identified earlier in these dykes. Possibly there is still a thermal influence from the young dyke outside the hybrid zone (>1.8 m from the contact) and this component may be an artifact combination of the "older" and "younger" components caused by overlapping of their Tub and coercivity spectra.

### 4.2. Locality 2

The Borg dyke carries a NRM (Fig. 6a; S2010.02) with a median destructive field (MDF, Dunlop and Özdemir, 1997) of ca. 15 mT. Three different components could be isolated, one erased in very low alternating field intensities (<20 mT) with a direction close to that of the present Earth field (PEF), a second of intermediate to high coercivity (decl. = 4.7°, incl. = 16.2°) and a third isolated after AF treatment in field higher than the previous components (>65 mT). The second component may be an artifact resulting from overlapping coercivity spectra expressed by the gradual change in direction along a great circle from the low to the highest coercivity direction. The mean remanence direction of the highest coercivity component (decl. = 181.5°, incl. = 28.9°, $\alpha_{95}$ = 14.4°; Table 2; Fig. 7b) is not

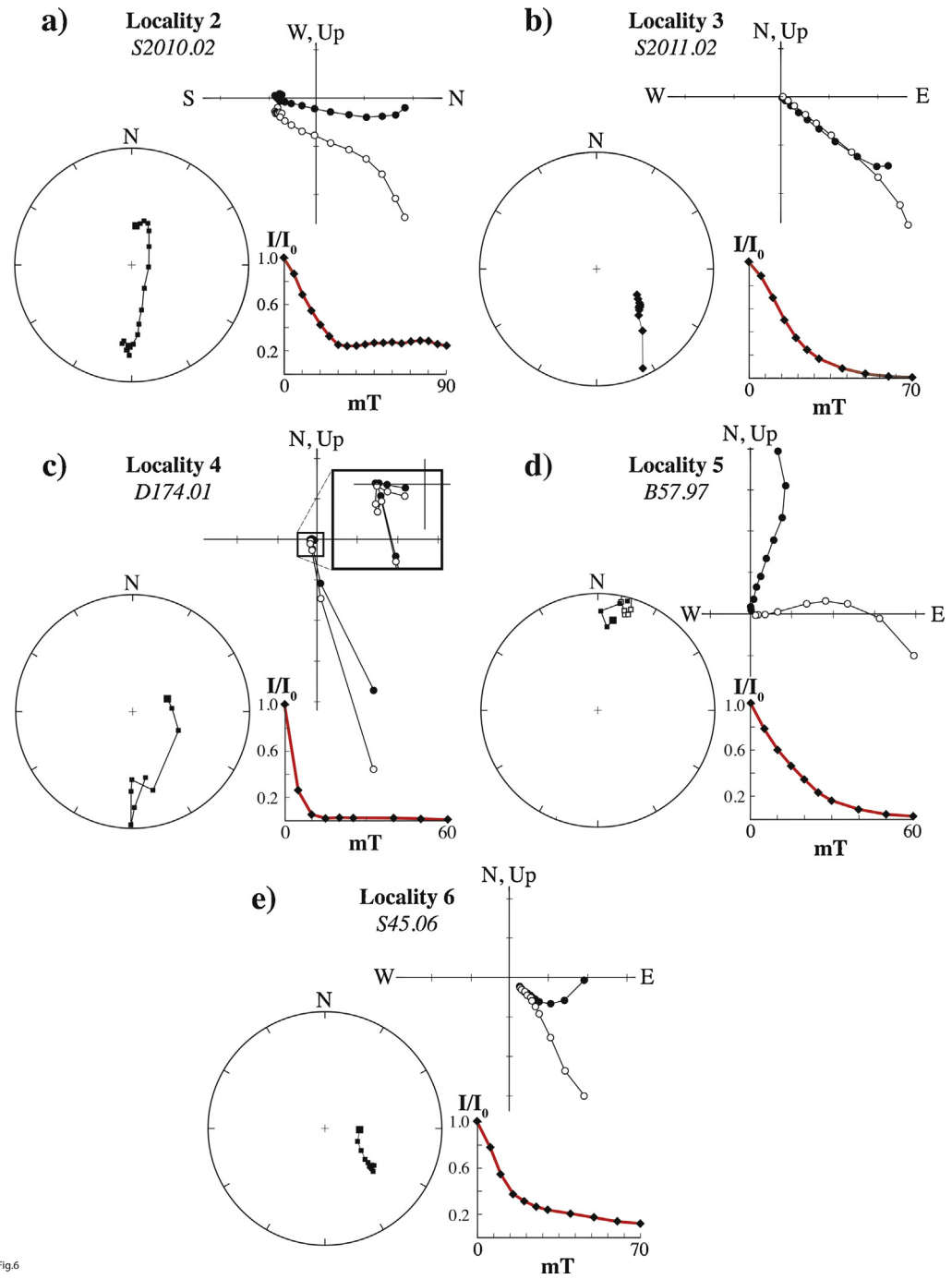



**Fig. 6.** Examples of remanence behavior during AF and thermal demagnetizations. In orthogonal plots, open (closed) symbols show magnetization vector endpoints in the vertical (horizontal) plane; curves show changes in intensity during demagnetization. Stereoplots show upwards (downwards) pointing palaeomagnetic directions with open (closed) symbols.

significantly different from the ChRM of the dykes O1 and O2 (O1a and O2a; Table 2; Fig. 7b).

### 4.3. Locality 3

The coercivity of the NRM carried by the Marbystrand dyke is rather low with a MDF of 13–20 mT (Fig. 6b; S2011.02). After removing a soft and poorly defined part of the remanence a stable component is isolated in AF higher than ca. 30 mT and a well defined mean ChRM direction is calculated (decl. = 127.5°, incl. = 31.1°, $\alpha_{95}$ = 3.6°; Table 2; Fig. 7a). Even though a stable remanence can be defined, the direction of remanence often changes along a great circle during AF demagnetization. Great circle analysis reveal a direction of intersecting great circles (decl. = 128.7°, incl. = 27.8°) supporting the mean calculated from the stable remanence directions. The mean ChRM direction is antipodal to the ChRM directions of dykes Y1 and Y2, i.e. the reversal test is positive with classification C (McFadden and McElhinny, 1990).

### 4.4. Locality 4

The NRM of the dyke in Rejmyre (1595 ± 3 Ma, Söderlund et al., 2005), which forms part of the Breven-Hällefors system of dykes, is of very low coercivity and >90% of the magnetization is lost

**Table 2**
New palaeomagnetic data from Proterozoic mafic dykes in southern-central Sweden.

| Locality-code/rock | Glat°/Glon° | *N*/*n* | Decl.° | Incl.° | K | α95° | Plat. N° | Plon. E° | $A_{95}$° | Age (Ma) | Comments | Trend° |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1/Dyke Y1 + BC | 58.58/16.33 | 1/18[a] | 311.3 | −43.9 | 73.0 | 4.1 | −3.4 | 239.0 | 4.0 | 939 ± 3 | | 38.0 E |
| *Dyke Y1* | | 1/13[a] | 311.2 | −43.3 | 53.7 | 5.7 | 3.0 | 239.3 | 5.6 | 939 ± 3 | | |
| *Baked contact (BC)* | | 2/5[a] | 311.5 | −45.4 | 584.1 | 3.2 | 4.5 | 238.4 | 3.3 | 939 ± 3 | | |
| 1/Dyke Y2 | 58.58/16.33 | 1/14[a] | 313.3 | −40.7 | 29.5 | 7.4 | 0.5 | 238.3 | 7.0 | ~as Y1 | | 38.0 E |
| *Hybrid (H) in O1 and O2* | | 2/11[a] | 344.3 | −22.8 | 30.9 | 8.3 | 18.4 | 212.5 | 6.4 | ~as Y1 | Tub = 300–530 °C | |
| 1/Dyke O1a (H + U) | 58.58/16.33 | 1/10[a] | 172.6 | 22.9 | 14.4 | 13.2 | 19.2 | 204.0 | 10.2 | ~as O2 | Tub = 500–580 °C | 23.0 W |
| *Hybrid in O1 (H)* | | 1/8[a] | 173.7 | 24.4 | 16.5 | 14.1 | | | | ~as O2 | Tub = 500–580 °C | |
| *O1 outside hybrid (U)* | | 1/2[a] | 168.5 | 16.9 | | | | | | ~as O2 | | |
| 1/Dyke O2a (H + U) | 58.58/16.33 | 1/11[a] | 172.8 | 19.7 | 14.7 | 12.3 | 21.0 | 203.9 | 9.3 | 1411 ± 9 | HHc/Htub | 23.0 W |
| *Hybrid in O2 (H)* | | 1/3[a] | 179.8 | 10.8 | 14.7 | 33.3 | | | | 1411 ± 9 | Tub = 520–590 °C | |
| *O2 outside hybrid (U)* | | 1/8[a] | 169.9 | 23.0 | 15.0 | 14.8 | | | | 1411 ± 9 | Tub ≥ 450 °C, HHc | |
| *O2 outside hybrid* | | 1/8[a] | 153.8 | −29.5 | 50.7 | 7.9 | 43.0 | 231.9 | 6.4 | 1411 ± 9 | Tub = 300–400 °C, Int Hc | 60.0 W |
| 2/Borg | 58.58/16.09 | 1/4[a] | 181.5 | 28.9 | 41.7 | 14.4 | 16.0 | 196.1 | 11.8 | | | 51.0 W |
| 3/Marbystrand | 58.61/16.48 | 1/9[a] | 127.5 | 31.1 | 202.9 | 3.6 | 3.3 | 246.0 | 3.0 | | Reversed | |
| 4/Rejmyre | 58.83/16.08 | 1/16[a] | 184.0 | 16.3 | 28.1 | 7.1 | 22.8 | 191.8 | 5.3 | 1595 ± 3[b] | 1 age[b] | E–W |
| 5/Hällefors | 59.14/16.48 | 1/6[a] | 21.9 | −19.7 | 20.1 | 15.3 | 18.5 | 173.7 | 11.6 | | | E–W |
| 6/Falun | 60.59/15.58 | 1/9[a] | 128.7 | 37.9 | 290.5 | 3.0 | 1.7 | 242.3 | 2.6 | 946 ± 1[b] | 1 age[b] | 28.0 W |
| 7/Mean Y1, Y2, Marby-strand | | 3[a]/41 | 310.6 | −38.6 | 131.7 | 10.8 | −0.1 | 241.1 | 9.9 | 939 ± 3 | | |
| 8/Mean O1a, O2a, Borg | | 3[a]/25 | 175.5 | 23.9 | 153.1 | 10.0 | 18.8 | 200.9 | 7.8 | 1411 ± 9 | | |

Glat/Glon = geographic latitude and longitude, *N*/*n* = number of sites/samples, Decl. (Incl.) = mean declination (inclination), K = Fischer precision parameter, α95 = semi-angle of the cone of 95% confidence, Plat. (Plon.) = latitude (longitude) of the (Virtual) palaeomagnetic poles, A95 = semi-angle of the 95% confidence circle of the pole position, age = radiometric age.

[a] Level of mean calculations.

[b] The ages are from Söderlund et al. (2005).

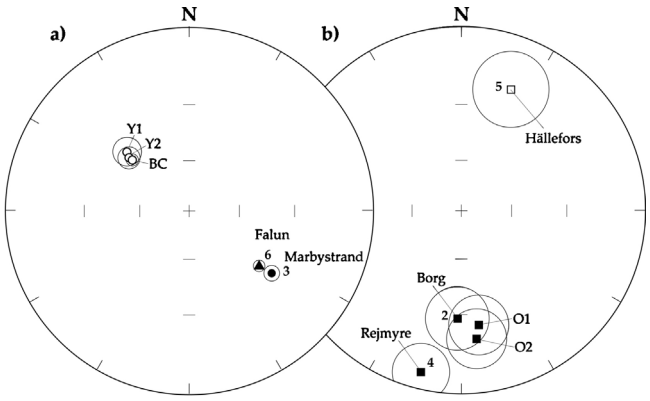



**Fig. 7.** Site mean remanence directions and corresponding 95% confidence circles for the basic dykes at (a) the locality 1, the dykes Y1, Y2 and baked contact, the locality 3 (Marbystrand), and the locality 6 (Falun). (b) The site mean direction from the older dykes at locality 1, O1 and O2, cross cut by Y1 and Y2, and the ca. 1595 Ma Rejmyre and Hällefors dykes (localities 4 and 5, respectively).

already at 10 mT demagnetizing fields. However, after removal of a soft remanence with a direction not far from that of the IGRF (decl. = 4.0°, incl. = 72.0°), a stable (MAD < 9°) and well defined direction (mean: decl. = 184.0°, incl. = 16.3°, $\alpha_{95}$ = 7.1°) is isolated in AF higher than 25 mT (Fig. 6c; 174.01). This direction is not very different from those of dykes O1, O2 in locality 1 and that of the Borg dyke in locality 2 (Fig. 7b), but it is also nearly antipodal to the ChRM direction of the 1577 ± 12 Ma Föglö Sottunga dyke (Pesonen and Neuvonen, 1981; Suominen, 1991. This ChRM has been isolated only during AF demagnetization The Tub spectra of the remanence in this dyke is very wide and from the thermal demagnetization a direction is identified (decl. = 65.9°; incl. = 65.8°, $\alpha_{95}$ = 14.0°), which is not significantly different from that of the low coercivity remanence (decl. = 51.6°, incl. = 67.2°, $\alpha_{95}$ = 7.7°).

### 4.5. Locality 5

The Hällefors dyke carries a remanence with a large portion of low coercivity (MDF ca. 10 mT; Fig. 6d; B57.97), but also rather well defined (MAD < 8°) high coercivity components that form a fairly well defined site mean (decl. = 21.9°, incl. = −19.7°, $\alpha_{95}$ = 15.3°). After changing the polarity of this site mean it is not significantly different from high coercivity and unblocking temperature components of the O1 and O2 dykes at locality 1 and from the site means in localities 2 and 4 (Fig. 7b).

### 4.6. Locality 6

The BDD dyke here has been dated at 946 ± 1 Ma (Söderlund et al., 2005). Well defined ChRM directions (MAD < 5°) are identified in this dyke after 40 mT AF demagnetizations (Fig. 6e). The site mean direction (decl. = 128.7°, incl. = 37.9°) is also well defined ($\alpha_{95}$ = 3.0°; Table 2; Fig. 7a) and have a polarity opposite to the directions of the ChRM of the Y1 and Y2 dykes at locality 1.

## 5. Origin of magnetization and anisotropy of magnetic susceptibility

The magnetization in the "young" dykes (Y1 and Y2) at locality 1 is carried by low-titanium titanomagnetite as indicated from unblocking temperatures (>ca. 530 °C) and susceptibility vs. temperature ($\kappa$–$T$) measurements (Curie temperature ca. 550 °C; Fig. 5a and 8a). This is true also for the baked contact of the "old"

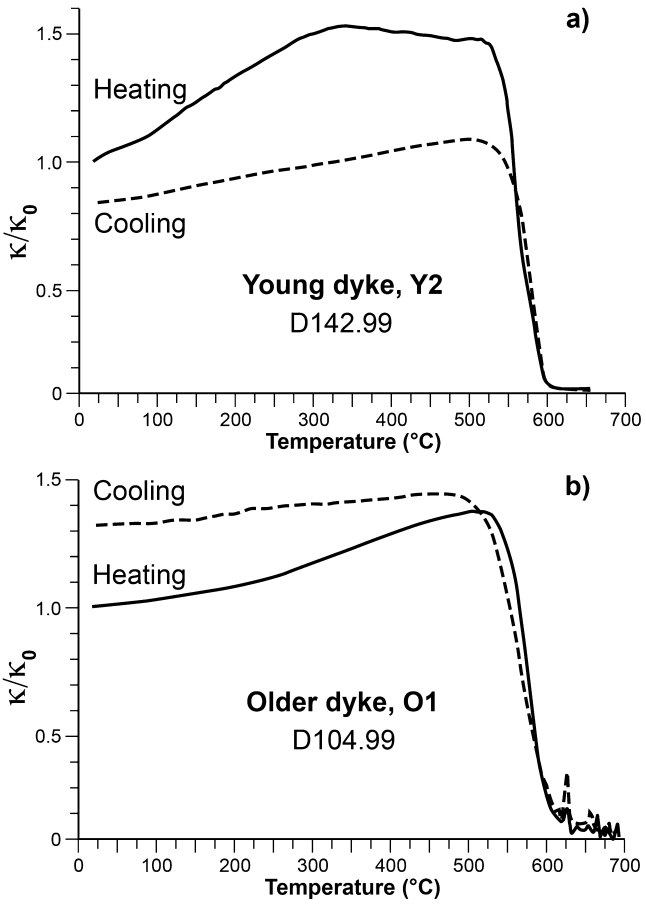



**Fig. 8.** Magnetic susceptibility during heating and cooling of samples from the Y2 and O2 dykes at site 1.

dyke O1 at the same locality, however, the Curie temperature indicates a somewhat higher titanium content (Fig. 8b). The presence of small, but noticeable, Hopkinson peaks in the $\kappa$–$T$ curves indicate that the remanence is carried in part by highly stable single-domain particles. Since single-domain particles carry stable remanent magnetizations, this gives an additional evidence for a primary magnetization.

The measurements of anisotropy of magnetic susceptibility (AMS) show a degree of anisotropy of 5–10% for the O1 and O2 dykes, 3–7% for the Y1 and Y2 dykes and less than 10% for the Falun dyke, which is the usual for igneous rocks with a primary magnetic fabric (Hrouda, 1982). The shape of the anisotropy ellipsoid is expressed by the shape parameter $T$ (Jelinek, 1981), for which $0 < T < 1$ express oblate shaped ellipsoid and $-1 < T < 0$ express a prolate ellipsoid. $T$ values are here positive ($T = 0.09$–$0.88$), indicating that the magnetic fabric for the dykes is oblate, i.e. the dykes are magnetically foliated. We interpret the AMS data as a primary flow indicator (Knight and Walker, 1988; Tauxe et al., 1998). The AMS data reveal orientations of the foliation planes of the young dyke Y1 (strike 29° E, dip 83°) that are parallel to the dyke plane. Also for the O1 and O2 dykes the magnetic foliation planes (strike 217° E, dip 46°) are parallel to the dyke planes, which are thus not vertical but dip ca. 45° to the southwest. This may suggest that these older dykes were initially inclined or that they have been tilted after intrusion.

The Falun dyke is also magnetically foliated, with a subvertical foliation plane (dip = 84° NE) trending 332° E, i.e. parallel to the trend of the dyke determined by geological mapping (SGU, 1964).

## 6. Discussion

### 6.1. New 945–932 Ma high quality pole for Baltica

The new 939 ± 3 Ma palaeomagnetic data from dykes Y1 and Y2 presented here is supported by rigorous baked contact test. There are several previously published ca. 970–910 Ma Baltican palaeopoles, some of them are precisely dated, but there were not a single one with proven primary ChRM (Table 3). However, the mean pole from just two dykes should be qualified as representing a near instantaneous recording of the Earth's magnetic field, and is not likely to have averaged palaeo-secular variation (PSV). To obtain a really valuable 'key' pole other palaeomagnetic results of similar age are needed.

Pisarevsky and Bylund (2006) published a palaeomagnetic study of the four 935 ± 3 Ma (U–Pb baddeleyite; Hellström et al., 2004) Göteborg and Slussen dykes carrying bipolar ChRM directions very similar to that of Y1 and Y2 dykes.

The 946 ± 1 Ma (U–Pb baddeleyite; Söderlund et al., 2005) Falun dyke carries almost the same ChRM direction (Locality 6; Plat = −1.7°, Plon = 242.3°, $A_{95}$ = 2.6°) as reversed Y1 and Y2 directions. It is also similar to the previous study of the same Falun dyke by Patchett and Bylund (1977; Plat. = −6.1°, Plon. = 237.6°, $A_{95}$ = 5.7°), however somewhat better defined. Considering that the remanence directions of Patchett and Bylund (1977) were not defined using vector analysis and the mean direction was calculated from least scatter in magnetic directions during demagnetization, which was the best approach at the time, we use the new pole presented here to represent the Falun dyke (locality 6, Table 2). Bylund and Elming (1992) and Bylund (1985) studied three dykes (Andersbro, Gällsjön and Ejen) located nearby and trending in the same direction as the Falun dyke. The ChRM directions are coherent and also very close to that of the Falun dyke Table 3; Fig. 7), which may argue for them being part of the same generation of dykes. The age difference between the Y1/Y2 dykes and BDD Falun dykes is relatively small (939 ± 3 and 946 ± 1 Ma).

Importantly, the Y1/Y2 dykes in Norrköping area and Falun dykes are located ca. 90–100 km east of the Protogine zone, i.e. much further to the east than has been suggested for the limit of crystal reworking during the Svecororwegian (Wahlgren et al., 1994). K–Ar ages of biotite from southeast Sweden (Åberg, 1978) indicate that the K–Ar system started to close at ca. 1.55 Ma and was completed at ca. 1.40 Ma, i.e. the rocks were not reheated above ca. 300 ± 50 °C, the closure temperature for K–Ar in biotite. Fission track studies (Tullborg et al., 1996) of subsidence and uplift from an area some 150 km south of the present study area, reveal temperatures of the present surface at 250 ± 50 °C at 820 ± 88 Ma. This resetting of the fission-track ages was interpreted related to sedimentary burial and uplift. This means that any tectonic disturbance or palaeomagnetic overprint related to the Sveconorwegian orogeny is unlikely.

As will be shown later, the drift velocity of Baltica at that time was apparently low, which makes it reasonable to combine the Y1/Y2, Göteborg-Slussen and Falun area data and here we present a mean 946–935 Ma pole, based on 10 different dykes: Plat. = −0.9°, Plon. = 240.7°, $A_{95}$ = 6.7° (Table 3). The reversal test of McFadden and McElhinny (1990) is positive with classification $C$, $\gamma/\gamma_{cr}$ = 5.6°/15.8°.

The ChRM direction of the Marbystrand dyke (locality 3) is also very close to the reversed Y1/Y2 direction (Fig. 7a). It is located only 2 km from them (Fig. 2), but has a significantly different trend, similar to the Breven-Hällefors swarm. However, the available palaeomagnetic directions from the latter are significantly different (Fig. 7b; Table 3) suggesting that the Marbystrand dyke may be related to the 939 ± 3 Ma magmatism. A combined pole based on eleven dykes (Y1, Y2, Marbystrand, Göteborg-Slussen and Falun

**Table 3**
Poles combined for the new 946–935 Ma key pole for Baltica.

| # | Used palaeopole | N/n | Plat.° | Plon.° | $A_{95}$° | Age (Ma) | Reference |
|---|---|---|---|---|---|---|---|
| 1 | Y1 dyke | 1/18[a] | −3.4 | 239.0 | 4.0 | 939 ± 3 | This study |
| 2 | Y2 dyke | 1/14[a] | −0.5 | 238.3 | 7.0 | 939 ± 3 | This study |
| 3 | Falun dyke | 1/19[a] | −1.7 | 242.3 | 2.6 | 946 ± 1 | This study; Söderlund et al. (2005) |
| 4 | Andersbo dyke | 1/7[a] | 1.0 | 229.0 | 7.3 | 946 ± 1 | Bylund and Elming (1992) |
| 5 | Gällsjön dyke | 1/5[a] | 18.0 | 241.0 | 8.5 | 946 ± 1 | Bylund and Elming (1992) |
| 6 | Ejen dyke | 1/8[a] | 6.0 | 248.0 | 6.7 | 946 ± 1 | Bylund (1985) |
| 7 | Tuve dyke | 1/23[a] | −14.0 | 237.9 | 3.7 | 935 ± 3 | Pisarevsky and Bylund (2006) and Hellström et al. (2004) |
| 8 | Small dyke | 1/3[a] | −13.9 | 235.8 | 8.8 | 935 ± 3 | Pisarevsky and Bylund (2006) and Hellström et al. (2004) |
| 9 | Hjuvik dyke | 1/12[a] | −3.3 | 246.9 | 10.3 | 935 ± 3 | Pisarevsky and Bylund (2006) and Hellström et al. (2004) |
| 10 | Slussen dyke | 1/6[a] | 3.2 | 248.5 | 9.2 | 935 ± 3 | Pisarevsky and Bylund (1998) |
| 11 | Mean 1–10 | 10[a]/115 | −0.9 | 240.7 | 6.7 | 946–935 | This study |
| 12 | Marbystrand dyke | 1/9[a] | 3.3 | 246 | 3 | ? | This study |
| 13 | Mean 1–10, 12 | 11[a]/124 | −0.5 | 241.2 | 6.1 | 946–935 | This study |

[a] Level of mean calculations.

area BDD dykes) is calculated (Table 3) and the reversal test is also positive with classification C, $\gamma/\gamma_{cr} = 4.1^\circ/15.6^\circ$. Both of these two combined poles (Table 3, entries 11 and 13) have quality factors (Q) of 7 (Van der Voo, 1990) and fulfill the criteria for a "key" palaeomagnetic pole. That is: (1) the rocks are well dated; (2) the number of samples is sufficient and the statistical demands a fulfilled; (3) the samples have been adequately demagnetized; (4) field test constrain the age of magnetization; (5) there is a structural control and tectonic coherence with craton involved; (6) there are both normal and reversed magnetizations; (7) there is no resemblance to palaeomagnetic poles of younger Neoproterozoic (e.g. Pisarevsky et al., 2008) or Phanerozoic (e.g. Torsvik et al., 2012) ages.

This combined pole also fulfills the criteria for a key palaeomagnetic pole used by Buchan et al. (2000), i.e. the age uncertainty is <±20 Ma, the remanence is properly isolated by AF and/or thermal demagnetizations, sufficient number of dykes to average out secular variations, the pole is demonstrated primary through field and reversal tests and dykes of similar age from a broad geographic area yield consistent poles.

### 6.2. New palaeopoles for 1595 ± 3 and 1411 ± 9 Ma

The Hällefors A (entry 37, Table 4; Piper, 1980) and locality 5 (Hällefors) poles most probably are from the same giant dyke. This dyke has a similar trend as the 1595 ± 3 Ma Rejmyre dyke (locality 4). ChRMs of two dykes are nearly antipodal (Table 2; Fig. 7b) and the palaeopoles are not significantly different from poles of the ca. 1570 Ma Kumlinge-Brändö and Föglö Sottunga dykes in Finland (Fig. 9; Table 4, entries 39–40). Hence we suggest that the Hällefors dyke belongs to the same magmatic event as the ca. 1595 Ma Rejmyre dyke. These two new VGPs are included to the Table 4 (entries 36 and 38).

Field observations and orientation of magnetic foliation plane indicate that the O1 and O2 dykes in site 1 are not vertical but dip ca. 45° to the southwest. Since there are no geological evidence for a post- 1411 ± 9 Ma and pre- 939 ± 3 Ma tilting, an initial inclined emplacement of the dyke is likely. The calculated pole positions (O1a: Plat. = 19.2°, Plon. = 204.0° and O2a: Plat. = 21.0°, Plon. = 203.9°, Table 2) are not significantly different from the pole calculated from the dyke at locality 2 (Borg) and a mean ca. 1411 Ma pole (entry 23 in Table 2; Fig. 9) is calculated from the O1a, O2a dykes and the Borg dyke (locality 2) poles. This mean pole plots not so far from the ca. 1450–1470 Ma reference poles (Fig. 9; Table 4, entries 24–33), and the 10–15° difference can be explained by 40–60 m.y. difference in age. Although only three dykes are included in the mean and secular variations are most probably not averaged out, this pole adds new data to the gap in reliable Baltica poles between ca. 1450 and 1270 Ma.

### 6.3. Sveconorwegian and Grenville loops and palaeogeographic implications

There is a strong geological evidence for the close position of Baltica and Laurentia during almost the whole Neoproterozoic (e.g. Winchester, 1988; Gee et al., 1994; Fairchild and Hambrey, 1995; Higgins et al., 2001). The match between the two continents was in part based on the apparent similarity between the so-called Grenville and Sveconorwegian APWP (Apparent Polar Wander Path) loops. There were debates about the exact shapes and the direction of younging of the poles (clockwise vs. anticlockwise loops; Fig. 10; e.g. Weil et al., 1998; Hyodo and Dunlop, 1993; Elming et al., 1993; Pisarevsky and Bylund, 1998), but there was a general agreement about two important features. First, the "bottoms" of both loops were marked with palaeopoles of ca. 980–930 Ma (the poles with highest southern latitudes). Second, the near-equatorial parts of both loops are populated by ca. 1000–1020 Ma poles and ca. 900–800 Ma poles.

The southern-most part of the Laurentian (Grenville) loop was constrained by palaeopole from the Haliburton Intrusion A (Buchan and Dunlop, 1976), with age estimated by authors at 980 ± 10 Ma.

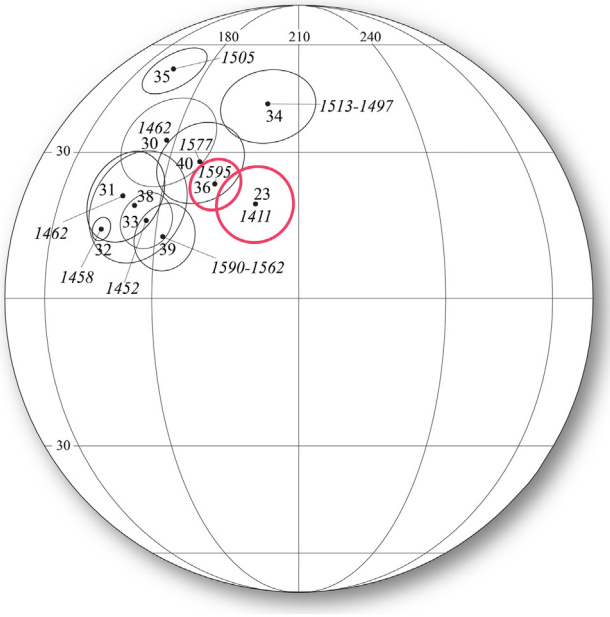



**Fig. 9.** Mean pole position (23) for the "old" 1411 Ma dykes O1, O2 and Locality 2 (Borg), together with the new 1595 Ma pole of Reymyre (36), both marked by the red confidence circle, and ca. 1400–1600 Ma reference poles for Baltica (pole numbers are as in Table 4).

**Table 4**
1600–850 Ma palaeomagnetic poles from Baltica.

| # | Pole | *N*/*n* | Plat.° | Plon.° | $A_{95}$° | Age (Ma) | Q | Reference |
|---|---|---|---|---|---|---|---|---|
| 9 | Hunnedalen dykes | 6[a]/79 | −41.0 | 222.0 | 10.5 | 848 ± 27 | 4 | Walderhaug et al. (1999) |
| 10 | Egersund-Onga anorthosite | 13[a]/67 | −42.1 | 200.4 | 9.0 | 900, 969 ± 14 | 3 | Brown and McEnroe (2004) and Walderhaug et al. (2007) |
| 11 | Rogaland Ingneous Complex | 5[a]/24 | −46.0 | 238.0 | 18.1 | 869 ± 14 | 3 | Walderhaug et al. (2007) |
| 12 | Egersund anorthosite | 76[a]/? | −43.5 | 213.7 | 3.6 | 936–929 | 4 | Walderhaug et al. (1999) and Stearn and Piper (1984) |
| 13 | Brattön-Älgö Norite-Anorthosite Complex | 11[a]/65 | 5.0 | 249.0 | 3.9 | 916 ± 11 | 3 | Stearn and Piper (1984) and Scherstén et al. (2000) |
| 14 | Mean 946–935 Ma Baltica pole | 10[a]/105 | −0.9 | 240.7 | 6.8 | 946–935 | 7 | This study |
| 15 | Tärnö-Fäjö dykes | 4[a]/21 | 12.8 | 246.7 | 15.5 | 946–954 | 3 | Patchett and Bylund (1992) and Söderlund et al. (2004, 2005) |
| 16 | Nilstorp dyke | 1/6[a] | 9.0 | 239.0 | 8.1 | 966 ± 2 | 3 | Patchett and Bylund (1992) and Söderlund et al. (2004) |
| 17 | ES metamorphic | 4[a]/47 | −45.9 | 219.4 | 5.9 | 980–930 | 3 | Pisarevsky and Bylund (1998) |
| 18 | Laanila Ristijärvi | 3[a]/54 | −2.1 | 212.0 | 16.5 | 1042 ± 50 | 3 | Mertanen et al., 1996 |
| 19 | Bamble Intrusion mean | 4[a]/ | 3.0 | 217.0 | 15.0 | 1100–1040 | 3 | Brown and McEnroe (2004) and Walderhaug et al. (2007) |
| 20 | Jämtland Complex | 16[a]/145 | −0.5 | 159.8 | 7.4 | ca. 1247 | 4 | Elming and Mattsson (2001) and Söderlund et al. (2006) |
| 21 | Västerbotten-Ulvö- | 32[a]/746 | −5.0 | 159.2 | 3.4 | 1259–1256 | 4 | Elming and Mattsson (2001) and Söderlund et al. (2006) |
| 22 | Dalarna Complex | 5[a]/53 | 5.9 | 160.4 | 10.2 | 1271–1264 | 4 | Elming and Mattsson (2001) and Söderlund et al. (2006) |
| 23 | Norrköping dykes | 3[a]/25 | 18.8 | 200.9 | 7.8 | 1411 ± 9 | 4 | This study (O1 + O2 + Borg dyke) |
| 24 | Bunkris | 1/8[a] | 27.0 | 197.0 | 2.8 | 1462 ± 1 | 3 | Bylund (1985) and Söderlund et al. (2005) |
| 25 | Glysjön | 1/6[a] | 37.0 | 180.0 | 5.7 | 1461 ± 1 | 3 | Bylund (1985) and Söderlund et al. (2005) |
| 26 | Söndagsåsen | 1/6[a] | 29.0 | 196.0 | 5.3 | 1461 ± 1 | 2 | Bylund and Elming (1992) |
| 27 | Emåsjön 13a | 1/7[a] | 39.0 | 173.0 | 7.7 | 1461 ± 1 | 2 | Bylund and Elming (1992) |
| 28 | Harpsjön 14a | 1/6[a] | 31.0 | 157.0 | 1.7 | 1461 ± 1 | 2 | Bylund and Elming (1992) |
| 29 | Noppikoski | 1/10[a] | 33.0 | 178.0 | 9.9 | 1461 ± 1 | 2 | Bylund and Elming (1992) |
| 30 | Mean Dalarna (24–29) | 6[a]/43 | 33.4 | 180.5 | 11.3 | 1461 | 4 | |
| 31 | Tuna dykes | 6[a]/38 | 20.6 | 170.5 | 9.5 | 1462–1461 | 4 | Piper (1992) |
| 32 | Valaam sill | 9[a]/36 | 13.8 | 166.4 | 2.4 | 1458 +4/−3 | 4 | Salminen and Pesonen (2007) |
| 33 | Ladoga dolerite dykes | 4[a]/12/278 | 15.2 | 177.6 | 5.5 | 1452 ± 12 | 6 | Lubnina et al. (2010) |
| 34 | Rödö basic dykes | 6[a]/53 | 41.6 | 201.7 | 9.5 | 1513, 1497 | 5 | Moakhar and Elming (2000) |
| 35 | Ragunda Fm | 15[a]/87 | 51.6 | 166.6 | 7.1 | 1505 | 4 | Piper (1979) |
| 36 | Rejmyre dyke | 1/16[a] | 22.8 | 191.8 | 5.3 | 1595 ± 3 | 3 | This study; Söderlund et al., 2005 |
| 37 | Hällefors A | 1/13[a] | 27.0 | 167.0 | 6.7 | As Rejmyre | 3 | Piper, 1980 |
| 38 | Hällefors B | 1/6[a] | 18.5 | 173.7 | 11.6 | As Rejmyre | 2 | This study; Söderlund et al., 2005 |
| 39 | Kumlinge-Brändö | 7[a]/25 | 12.2 | 182.0 | 6.7 | 1562–1590 | 4 | Pesonen and Neuvonen (1981) and Suominen (1991) |
| 40 | Föglö Sottunga | 6[a]/25 | 27.8 | 187.5 | 9.0 | 1577 ± 12 | 4 | Pesonen and Neuvonen (1981) and Suominen (1991) |

Pole numbers start from 9 in continuation to numbering in Table 2. *Q* = quality index (Van der Voo, 1990). See also footnotes to Table 2.

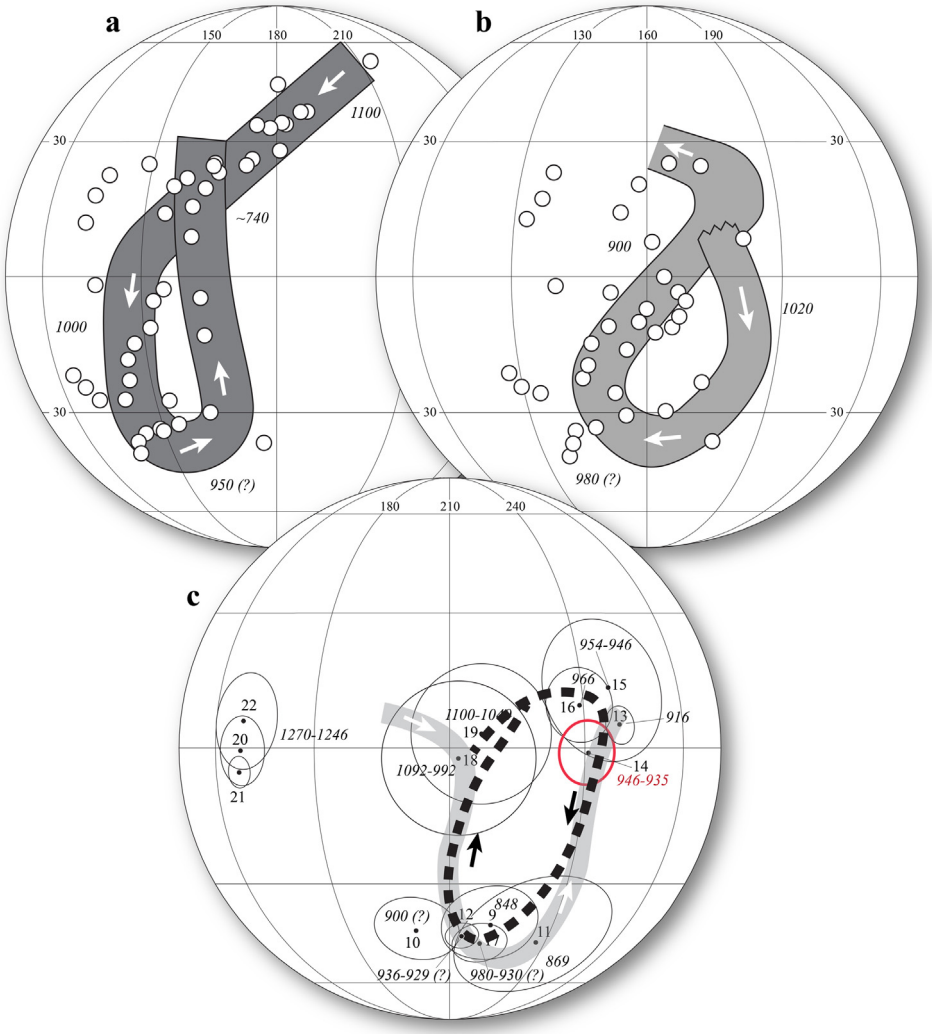



**Fig. 10.** The Grenvillian loop from (a) Weil et al. (2006; selected "A-poles" from Weil et al., 1998; Warnock et al., 2000) and (b) Hyodo and Dunlop, (1993; data from their Table 5). (c) The "Sveconorwegian" loop for Baltica in gray (e.g. Elming et al., 1993) and the new clockwise loop (hatched line) based on the key-pole presented here. Pole numbers are as in Table 4.

Warnock et al. (2000) restudied this intrusion both geochronologically and palaeomagnetically and demonstrated, that the best estimate of the age of this magnetization is ca. 30–40 m.y. older, between 1030 and 1000 Ma.

The southernmost part of the Baltica (Sveconorwegian) loop was constrained by several studies of the ca. 930 Ma Rogaland Igneous Complex (RIC) and related rocks (e.g. Table 4, entries 10–12). These poles are in an apparent contradiction with the new key pole, which is presented in this study, and all of them revealed steep upwards ChRM directions. Brown and McEnroe (2004) suggested a significant difference between the time of intrusion and the age of ChRM, suggesting the latter to be ca. 900 Ma. Walderhaug et al. (1999) presented a well-dated 848 Ma primary pole from the Hunnedalen dykes, which was very close to the poles from the Rogaland complex. Lately Walderhaug et al. (2007) obtained a $869 \pm 14$ Ma $^{40}Ar/^{39}Ar$ biotite age from a noritic dyke related to RIC and argued that it represents an uplift/cooling age, which better represents the age of the remanence in SW Norway. Palaeomagnetic studies of mostly undated >60 mafic dykes in the Protogine Zone in southern Sweden revealed that ChRM of all dykes south of the Ljungby – Alvesta road, i.e. south of the 57° N latitude, is upward steep, like the RIC ChRM. Most of dykes north of this line are magnetized with more shallow inclinations, similar to the BDD dykes in the Falun area. Pisarevsky and Bylund (1998) also found RIC-like remanence direction in the ca. 950–930 Ma metamorphic rocks in southern Sweden west of the Protogine Zone. Such geographically bound distribution of ChRM directions suggests a regional remagnetization event. The timing of this remagnetization is probably constrained by the ca. 870–850 Ma poles of Walderhaug et al. (1998, 2007), but the mechanism is unclear. Page et al. (1996) reported a detailed $^{40}Ar/^{39}Ar$ study in the Eastern Segment of the Sveconorwegian orogen in southern Sweden. They did not find any ages younger than 915 Ma. Hence the ca. 870–850 Ma remagnetization event in southern Sweden, if occurred, should have been of a low-temperature nature. New studies are necessary.

We agree with Walderhaug et al. (1998, 2007) and suggest that the real age of the southernmost palaeopoles constraining the Sveconorwegian loop is likely to be ca. 870–850 Ma. If so, the Grenville and Sveconorwegian loops are not coeval – their age difference is ca. 150 Ma (Fig. 11). However, most poles used for the Grenville loop were from poorly dated metamorphic rocks with uncertain age of their remanence. Using only well-dated Laurentian poles, only the older, "downward" part of the Grenville loop is palaeomagnetically constrained and there are no reliable Laurentian poles between 1000 and 790 Ma (Fig. 11; Table 5). On the other hand, Sveconorwegian loop is poorly defined between ca. 1100 and 1000 Ma

**Table 5**
Ca. 1090–1000 Ma Laurentian palaeomagnetic poles.

| Abbr | Rockname | Age (Ma) | Plat (°N) | Plong (°E) | $A_{95}$ (°) | Q | Reference |
|---|---|---|---|---|---|---|---|
| LST | Lake Shore Traps | 1089–1085 | 22.2 | 180.8 | 4.5 | 5 | Diehl and Haig (1994) and Davis and Paces (1990) |
| FS | Freda Sandstone | 1080–1020 | 2.2 | 179.0 | 4.2 | 4 | Henry et al. (1977) and Wingate et al. (2002) |
| NS | Nonesuch Shale | 1080–1020 | 7.6 | 178.1 | 5.5 | 4 | Henry et al. (1977) and Wingate et al. (2002) |
| CS | Chequamegon Sandst. | 1050–990 | −12.3 | 177.7 | 4.6 | 4 | McCabe and Van der Voo (1983) |
| JS | Jacobsville Sandstone | 1050–990 | −10.0 | 184.0 | 4.2 | 5 | Roy and Robertson (1978) |
| HI | Haliburton Intrusion | 1030–1000 | −32.6 | 141.9 | 6.3 | 3 | Warnock et al. (2000) |

Abbr = abbreviations used in Fig. 11; see also footnotes to Table 2. *Q* = quality index (Van der Voo, 1990).

by just two poles with very imprecise ages (Table 4, entries 18–19), but better between ca. 960–840 Ma (Fig. 11).

In spite of this incompatibility of the two APWP loops, we still suggest the integrity of Laurentia and Baltica in the Neoproterozoic as it is shown in Fig. 12. First, there is a geological evidence for this reconstruction after ca. 1000 Ma (e.g. Li et al., 2008; Pisarevsky et al., 2008; Cawood et al., 2010 and references therein). In particular, it is consistent with the existence of the long-lived Meso- to Neoproterozoic passive continental margin along the eastern and northeastern edge of Baltica. The eastern Greenland was also a passive continental margin during most of Neoproterozoic (Li et al., 2008 and references therein), which is consistent with the reconstruction in Fig. 12. On the contrary, outcrops of Neoproterozoic sediments in western Scandinavia are fragmented and generally interpreted to represent non-marine environments in the middle of Laurentia-Baltica continent (Fig. 12; e.g. Winchester, 1988). Second, the coeval ca. 615 Ma poles from Laurentia and Baltica (Murthy et al., 1992; Kamo and Gower, 1994; Walderhaug et al., 2007) support this reconstruction for the Early Ediacaran time and match the suggested model of the Late Ediacaran opening of the Iapetus ocean (e.g. Pisarevsky et al., 2008).

Fig. 12 shows the palaeomagnetically permissive reconstruction of Laurentia and Baltica (the central part of Rodinia) at 940 and 870 Ma. The closeness of ca. 960–920 Ma poles (Table 4; Fig. 10) suggests little movement of Laurentia – Batica in low latitudes (Fig. 12a). After ca. 910 Ma there was a quick southward movement

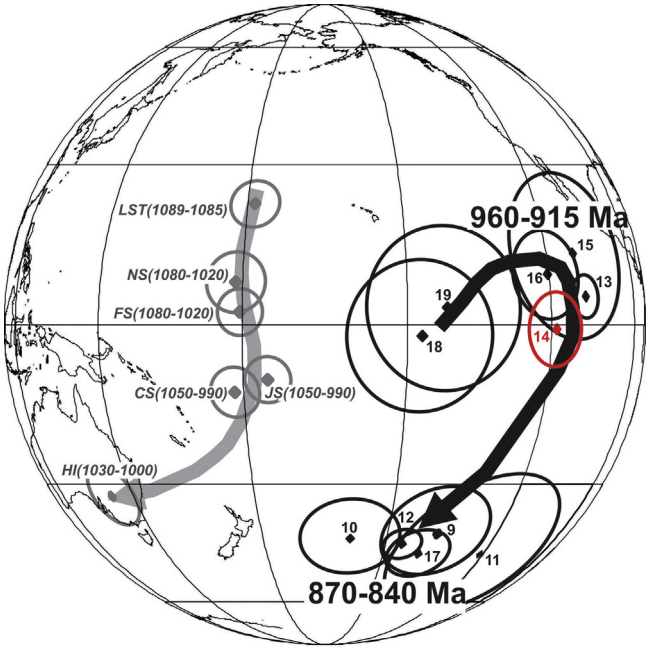



**Fig. 11.** APWPs for Laurentia (gray) and Baltica (black) that form a part of the Grenville and Sveconorwegian loops. Note the age difference in the palaeomagnetic poles forming the two APWP. Baltican pole numbers are as in Table 4, Laurentian poles are marked as in Table 5.

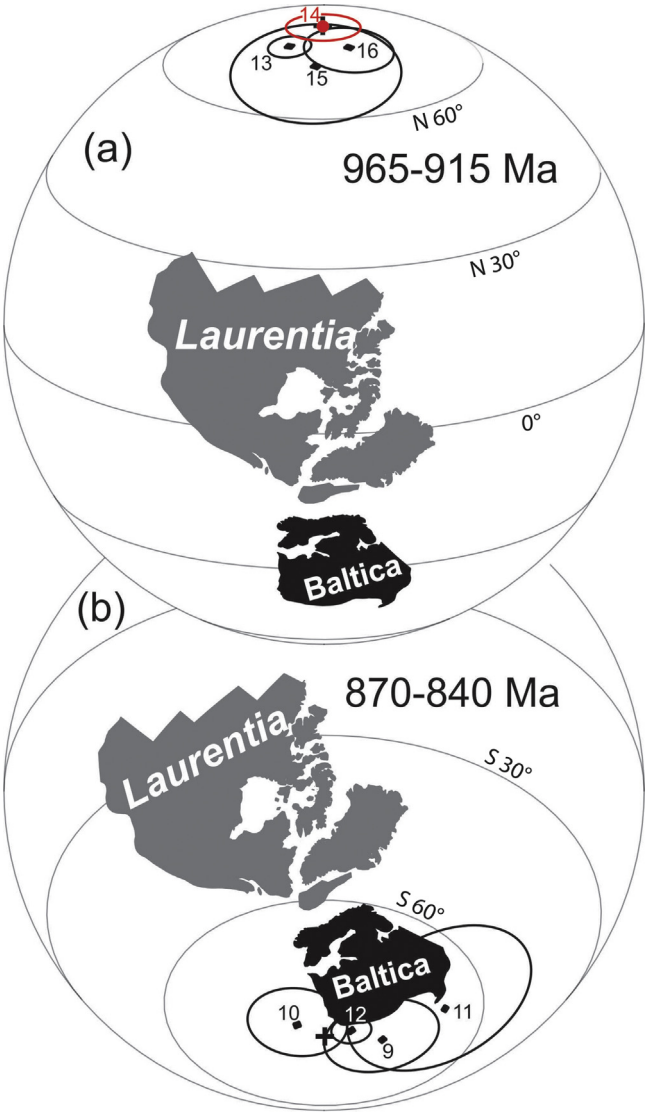



**Fig. 12.** The reconstructed positions of Laurentia-Baltics at: (a) ca. 965–915 Ma (Laurentia rotated to the absolute framework about a pole at 33.1°N, 120.0°W, −101.0°; Baltica is rotated to Laurentia at 75.8°N, 95.7°W, −59.2°) and (b) ca. 870–840 Ma (Laurentia rotated to the absolute framework about a pole at 19.6°N, 156.5°W, −130.1°; Baltica is rotated to Laurentia as in (a)). Poles are numbered as in Table 4. Note the big change in latitudinal positions between the two time intervals.

of the two, which resulted in the near-polar position of Baltica at ca. 870–850 Ma.

## 7. Conclusions

We obtained the first age determinations for mafic dykes in the southeastern part of the southern Svecofennian Subprovince.

The palaeomagnetic and geochronological study of the 939 ± 3 Ma mafic dykes near Norrköping, Sweden revealed a primary remanence supported by a rigorous contact test.

Averaging this and previously published palaeomagnetic results on coeval mafic dykes in other areas provide a new 946–935 Ma key pole for Baltica. This primary pole is significantly different from the palaeopole of the ca. 930 Ma Rogaland Igneous Complex (RIC) and some coeval mafic and metamorphic rocks in southern Sweden. We support the previous suggestion of Walderhaug et al. (2007), that the remanence of RIC-related rocks in southern Norway is significantly younger than the time of the igneous event. We also suggest a low-temperature remagnetization event in southern Sweden after ca. 900 Ma.

The new 946–935 Ma pole indicates that the distribution of poles in the "Sveconorwegian loop" was clockwise as has also been suggested for the Grenville loop of Laurentia.

Palaeogeographic reconstructions demonstrate a slow low latitudinal drift of joint Laurentia – Baltica continents between ca. 960 and 920 Ma and a quick movement towards the polar area by ca. 870–850 Ma.

The two new palaeopoles of ca. 1411 Ma and ca. 1595 Ma added to the palaeomagnetic database give new constrains to the apparent polar wander of Baltica.

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